DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I (19)

Subject : Operating Systems

Class: B.E (CSE)
Semester: IV
Academic Year: 2025-2026

Max Marks: 30
Duration:1 ½ hr
Date : 26-03-2026

S. No.	PART A Answer All Questions (Q5 x 2M = 10 M)	Marks	BT	CO
1.	Consider a Parent process P that has forked a child process C, now P terminates while C is still running. Answer yes/ no with a brief justification for the following question Will C immediately become a Zombie?	3	3	1
2.	Differentiate between dispatcher and scheduler.	2	3	1
3.	Give the objective of multiprogramming and time sharing.	2	2	1
4.	Define **graceful degradation** and **high availability** in the context of clustered systems. Which types of systems typically require these properties?	3	1	1
	PART B **Answer Any Two Questions (02 x 10M = 20M)**			
5.	Consider the following set of processes, with the length of the CPU burst given in milliseconds: Low 2 High (table below) The processes are assumed to have arrived in the order P1,P2,P3, P4,P5 all at time 0. (i) Draw four Gantt charts that illustrate the execution of these processes using FCFS, SJF, non preemptive priority (ii) What is the turn around time of each process for each algorithm (iii) What is the waiting time of each process for each algorithm calculate Average waiting time	10.	3	1
6.	a) Justify the statement "Operating system can be viewed as a government, resource allocator and a control program".	6	3	1
	b) What is the use of medium term scheduler.	4	2	1
7.	a) Compare shared memory and message passing in terms of speed, ease of implementation, scalability, risk of data inconsistency	5	3	1
	b) What is cache coherency problem? In what type of system does it commonly arise	5	1	1

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3